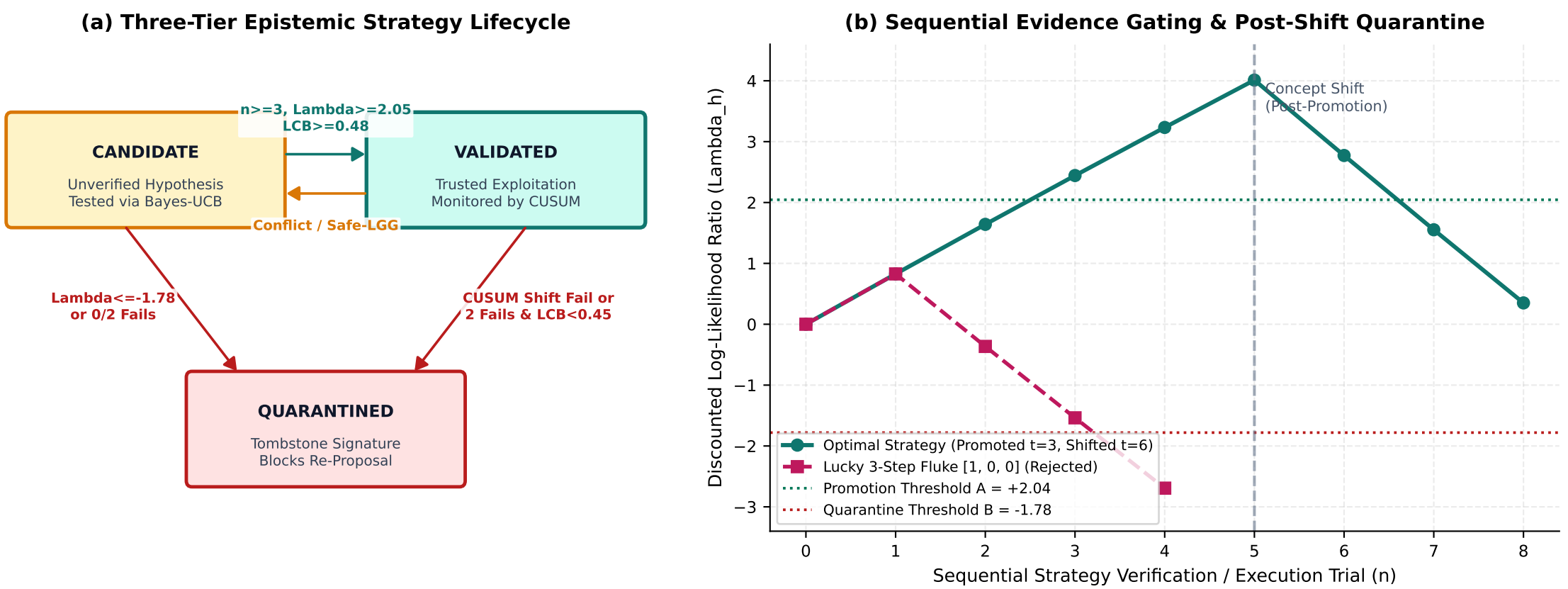
(a) Three-Tier Epistemic Strategy Lifecycle
n>=3, Lambda>=2.05
LCB>=0.48
CANDIDATE
Unverified Hypothesis
Tested via Bayes-UCB
VALIDATED
Trusted Exploitation
Monitored by CUSUM
Conflict / Safe-LGG
Lambda<=-1.78
or 0/2 Fails
CUSUM Shift Fail or
2 Fails & LCB<0.45
QUARANTINED
Tombstone Signature
Blocks Re-Proposal
(b) Sequential Evidence Gating & Post-Shift Quarantine
Concept Shift
(Post-Promotion)
Optimal Strategy (Promoted t=3, Shifted t=6)
Lucky 3-Step Fluke [1, 0, 0] (Rejected)
Promotion Threshold A = +2.04
Quarantine Threshold B = -1.78
Discounted Log-Likelihood Ratio (Lambda_h)
Sequential Strategy Verification / Execution Trial (n)
4
3
2
1
0
−1
−2
−3
0
1
2
3
4
5
6
7
8